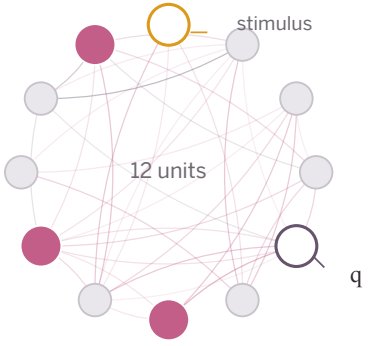
Predictive risk and recording decisions

Does prior averaging improve neural stimulus-response prediction?

Known connectivity

Uncertain neuronal physiology



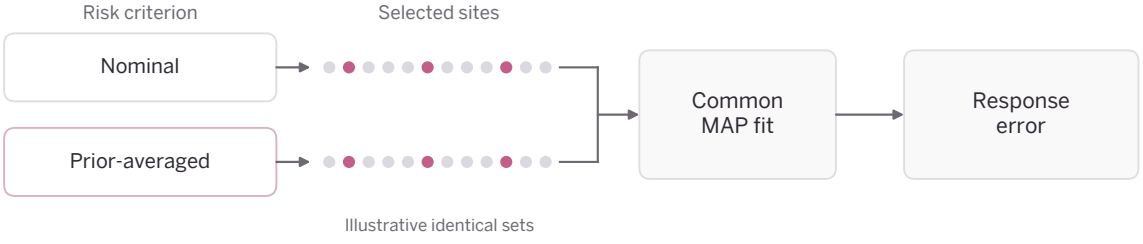
- 3 recorded units
- stimulus
- unrecorded readout

J5000: fixed example network

Connections: rose + / plum

Recording decisions

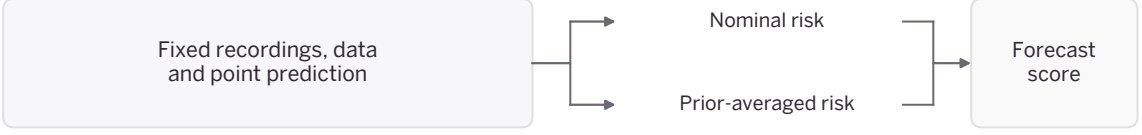
40 independent networks



Same recordings imply identical point predictions

Risk forecasts

40 new independent networks



Often the same recordings

82/120 single-target designs

26/40 shared designs

Identical nominal / prior-averaged sets

Mean-MSE change (%)

`-3.7% [-18.5, +12.1]`

`+2.7% [ -3.0, +11.5]`

Single / shared objective · 97.5% CI

Forecast score

`-0.46 [-1.16, +0.02]`

95% CI · mean influenced by one network

All effects: prior-averaged minus nominal

Primary improvements remain uncertain; decisions and forecasts require separate evaluation.

Jiayi Du & Shuaizhi Cheng · Equal contribution

Simulation study · known connectivity